

ISFAHAN SHAHID BEHESHTI INTL, Iran, Islamic Republic of

WMO: 408000

Lat: 32.751N Lon: 51.862E Elev: 5072 StdP: 12.19 Time Zone: 3.50 (IRN) Period: 94-19 WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 17.2	(c) 19.8	(d) -0.7	(e) 6.4	(f) 39.1	(g) 2.8	(h) 7.7	(i) 40.6	(j) 25.8	(k) 54.4	(l) 22.1	(m) 50.2	(n) 3.5	(o) 270	(p) 0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.0	102.4	62.2	100.6	61.3	98.6	60.7	65.4	96.8	63.7	95.5	62.4	94.4	7.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.3	73.0	78.0	50.2	65.1	72.8	48.2	60.3	72.0	33.2	96.8	31.7	95.4	30.6	94.4	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.0	(b) 19.1	(c) 15.9	(e) 11.0	(f) 105.0	(g) 3.6	(h) 1.4	(i) 8.5	(j) 106.0	(k) 6.4	(l) 106.8	(m) 4.4	(n) 107.6	(o) 1.8	(p) 108.6		
			DB													
			WB													
				9.4	67.9	3.1	3.7	7.2	70.5	5.4	72.7	3.6	74.7	1.4	77.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.1	36.5	41.8	50.4	59.4	69.2	78.8	83.5	79.6	72.6	61.8	47.1	39.3
	DBStd	16.93	6.00	6.40	6.35	5.49	4.91	4.04	3.89	3.85	3.94	4.76	6.30	5.39
	HDD50	1196	419	237	73	3	0	0	0	0	0	1	129	334
	HDD65	3641	883	651	453	179	18	0	0	0	2	121	537	797
	CDD50	4885	1	6	86	286	596	865	1037	919	679	366	42	2
	CDD65	1856	0	0	1	12	149	415	572	454	231	22	0	0
	CDH74	30853	0	0	47	359	2531	6615	9349	7360	3800	787	5	0
	CDH80	17631	0	0	6	55	1023	3927	6020	4504	1924	172	0	0
(14) Wind	WSAvg	6.5	5.1	6.7	7.7	8.0	8.2	7.5	6.9	6.3	6.0	5.8	4.8	4.6
(15) Precipitation	PrecAvg	4.8	0.9	0.6	1.0	0.6	0.3	0.0	0.0	0.0	0.0	0.2	0.4	0.7
	PrecMax	7.7	2.6	1.8	2.6	1.5	1.1	0.4	0.4	0.1	0.1	1.4	1.5	1.6
	PrecMin	2.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.5	0.6	0.5	0.7	0.5	0.3	0.1	0.1	0.0	0.0	0.3	0.5	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	68.0	80.3	85.4	93.6	102.3	104.4	103.6	95.4	87.5	73.7	64.3
		MCWB	43.2	45.8	50.6	54.3	57.8	61.3	63.3	61.8	58.8	55.5	50.0	46.0
	2%	DB	57.3	62.9	73.6	80.9	91.2	100.3	103.7	100.6	94.6	84.3	69.9	59.2
		MCWB	41.2	43.5	47.8	53.3	56.8	60.6	63.0	61.6	58.7	55.0	48.5	43.3
	5%	DB	53.8	60.4	69.8	78.5	88.2	98.3	102.1	98.6	92.5	82.0	66.3	55.5
		MCWB	40.0	42.6	46.6	52.2	56.2	59.7	62.2	61.2	57.8	54.1	47.4	41.4
	10%	DB	50.2	57.0	66.3	75.2	86.1	95.4	100.2	96.7	89.9	78.8	62.5	52.1
		MCWB	38.1	40.9	45.4	50.6	55.5	58.5	61.2	60.7	57.0	52.8	45.7	40.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.2	48.5	52.6	57.0	62.3	66.4	68.6	67.7	62.9	58.4	52.5	47.7
		MCDB	55.4	61.6	66.7	75.1	86.8	96.9	98.4	95.0	91.1	81.8	67.5	57.6
	2%	WB	43.6	46.0	50.3	55.2	59.4	63.6	65.9	64.5	60.6	56.3	50.4	45.7
		MCDB	53.1	58.2	67.1	74.1	86.2	94.2	98.1	95.6	90.0	80.1	64.1	54.9
	5%	WB	41.3	44.1	48.7	53.8	57.9	61.5	64.2	62.9	59.0	54.9	49.0	43.8
		MCDB	50.8	56.5	65.2	72.9	83.9	92.9	96.9	95.4	88.6	77.7	62.3	52.1
	10%	WB	39.3	42.0	46.9	52.4	56.5	59.9	62.8	61.6	57.7	53.4	47.2	41.8
		MCDB	48.5	54.6	63.1	70.8	81.3	91.3	95.4	94.1	87.3	75.4	60.2	49.6
(35) Mean Daily Temperature Range	MDBR		25.7	28.0	28.9	28.1	30.8	32.6	32.0	34.3	34.7	32.3	27.7	25.5
	5% DB	MCDBR	30.5	31.8	33.2	31.7	33.6	35.2	33.5	34.8	36.1	35.0	32.2	30.2
		MCWBR	18.3	17.5	16.5	13.8	13.4	13.8	12.9	14.3	15.6	16.7	17.4	17.9
	5% WB	MCDBR	26.1	27.2	27.9	27.5	31.2	32.9	30.6	33.1	34.4	32.3	27.3	24.7
		MCWBR	16.6	16.3	14.8	12.6	13.0	14.1	12.9	14.6	15.9	16.0	15.3	15.7
(40) Clear-Sky Solar Irradiance	taub		0.290	0.325	0.384	0.463	0.467	0.455	0.480	0.432	0.382	0.381	0.329	0.290
	taud		2.447	2.319	2.108	1.915	1.924	1.967	1.921	2.052	2.179	2.156	2.333	2.463
	Ebn at Noon		297	294	282	264	263	265	258	269	279	268	277	290
	Edh at Noon		29	36	48	61	61	58	61	52	44	42	32	27
(44) All-Sky Solar Radiation	RadAvg		1054	1363	1740	1961	2300	2628	2505	2372	2044	1576	1142	972
	RadStd		69	84	95	126	117	81	67	55	52	65	85	78

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-2.00	N/A	N/A	-2.23	N/A	N/A	+207	-176	-110
Regional (3 neighbors)	N/A	N/A	N/A	N/A	-1.78	N/A	N/A	+207	-176	-110

Nomenclature: See separate page